UNIT - 5
MODEL FITTING IN R

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- Regression Models- Linear and Logistic Model
- Classification Models– Decision Tree
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Regression Models

Regression is one of the most important techniques in Machine Learning and Data Science. It is used to understand the **relationship between input variables and an output variable** and to make predictions.

There are two commonly studied regression models:

1. **Linear Regression**
2. **Logistic Regression**

Linear Regression

It is a **supervised machine-learning algorithm** used to predict a **continuous numerical value** based on one or more input variables.

Linear Regression is a supervised machine learning and statistical technique used to study the relationship between a **dependent variable (Y)** and one or more **independent variables (X)** and to predict the value of Y.

For simple linear regression, the basic equation is:

$$Y = b0 + b1\ X$$

Linear Regression

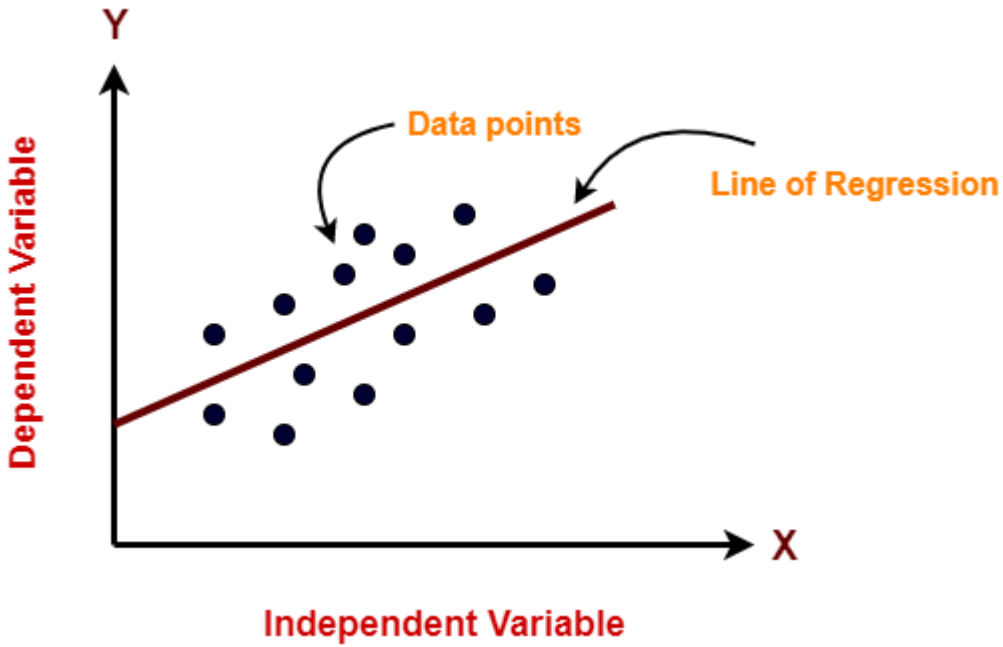
Where:

Y = predicted/dependent variable

X = independent variable

$\mathbf{b_0}$ = intercept

$\mathbf{b_1}$ = slope/coefficient



Linear Regression

Examples:

1. Predicting house rent
2. Predicting salary
3. Predicting temperature
4. Predicting sales
5. Predicting student's marks
6. Predicting product demand

Why Do We Use Linear Regression?

Linear regression is mainly used for **prediction** and **understanding relationships** between variables.

Example 1 – House Price

We can use:

X = House size (sq. ft.)

Y = House price

The model can learn the relationship between **house size** and **price** and **predict the price of a new house.**

Why Do We Use Linear Regression?

Example 2 – Student Marks

X = Hours studied

Y = Marks

We can predict approximately how many marks a student may obtain based on study hours.

Example 3 – Sales Prediction

X = Advertising expenditure

Y = Sales

It can help estimate future sales based on advertising spending.

Importance of Linear Regression

1. **Prediction**

   It helps predict unknown or future values.

2. **Understand Relationships**

   It tells us how one variable is related to another.

3. **Easy to Understand**

   The model is relatively simple and easy to interpret.

4. **Decision Making**

   Businesses can use predictions to make better decisions.

Importance of Linear Regression

5. Trend Analysis

It helps identify whether a variable is increasing or decreasing with another variable.

6. Data Analysis

It is widely used in statistics, economics, business, healthcare and machine learning.

Advantages of Linear Regression

1. Simple and Easy

It is easy to understand and implement.

2. Fast

It requires relatively less computational power.

3. Easy Interpretation

The coefficients clearly indicate how changes in X are associated with changes in Y.

4. Good for Linear Relationships

It works well when the relationship between variables is approximately linear.

Advantages of Linear Regression

5. Useful for Prediction

It can provide useful predictions when its assumptions are reasonably satisfied.

6. Works with Small Datasets

It can often be trained effectively even when the dataset is relatively small.

Disadvantages of Linear Regression

1. Assumes Linear Relationship

It performs poorly when the relationship between X and Y is strongly nonlinear.

2. Sensitive to Outliers

A few unusual observations can significantly change the regression line.

3. Multicollinearity Problem

When independent variables are highly correlated with each other, coefficient estimates can become unstable.

Disadvantages of Linear Regression

4. Sensitive to Irrelevant Variables

Including unnecessary variables can reduce model quality and make interpretation difficult.

5. Not Suitable for Every Problem

It may not work well for complex patterns that require nonlinear or more flexible models.

Types of Linear Regression

A. Simple Linear Regression

B. Multiple Linear Regression

A. Simple Linear Regression

It has **one independent variable (X)** and **one dependent variable (Y)**.

$$Y = b0 + b1\ X$$

Example:

Predicting salary based on years of experience.

X = Years of experience

Y = Salary

Types of Linear Regression

B. Multiple Linear Regression

It has **two** or **more independent variables (X)** and **one dependent variable (Y)**.

$$Y = b0 + b1\ X1 + b2\ X2 + b3\ X3 + \cdots + bn\ X$$

Example:

Predicting house price based on:

X_1 = House area

X_2 = Number of bedrooms

X_3 = Location score

Y = House price

Logistic Regression

Logistic Regression is a **supervised machine learning algorithm** mainly used for **classification problems**. It predicts the probability that an observation belongs to a particular class.

For example:

- ✓ **Email** → Spam / Not Spam
- ✓ **Patient** → Disease / No Disease
- ✓ **Student** → Pass / Fail
- ✓ **Customer** → Buy / Not Buy

Unlike linear regression, which predicts a continuous numerical value, logistic regression generally predicts a **class** or **category**.

Why Do We Use Logistic Regression?

We use logistic regression when the **output is categorical**, especially when we want to predict the probability of an event.

Example: Disease Prediction

Suppose we want to predict whether a patient has a disease.

X = Age, blood pressure, sugar level, etc.

Y = Disease / No Disease

Logistic regression can calculate a probability such as:

Probability of disease = 0.82 (82%)

If the selected threshold is 0.5, the patient would be classified as **Disease = Yes**.

Importance of Logistic Regression

1. Classification

It is widely used to classify data into different categories.

2. Probability Prediction

It provides a probability between **0 and 1**, which is useful for decision-making.

3. Simple and Interpretable

The model is relatively easy to understand compared with many complex machine-learning algorithms.

Importance of Logistic Regression

4. Feature Analysis

Its coefficients can help understand how input variables are associated with the predicted outcome.

5. Widely Used

- Healthcare
- Banking
- Marketing
- Education
- Fraud detection
- Email spam detection

Advantages of Logistic Regression

1. **Easy to Implement**

   It is relatively simple to build and train.

2. **Fast**

   It is computationally efficient for many classification problems.

3. **Gives Probabilities**

   It provides estimated probabilities, not just class labels.

4. **Easy to Interpret**

   The effect of input variables can be analyzed using model coefficients and odds ratios.

Advantages of Logistic Regression

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4. **Easy to Interpret**

   The effect of input variables can be analyzed using model coefficients and odds ratios.

Advantages of Logistic Regression

5. Works Well with Linear Decision Boundaries

It performs well when the classes can be reasonably separated by a linear boundary in the feature space.

6. Supports Regularization

Techniques such as **L1 and L2 regularization** can help control overfitting.

Disadvantages of Logistic Regression

1. **Linear Decision Boundary**

   Basic logistic regression may not perform well when the relationship between features and classes is highly nonlinear.

2. **Sensitive to Outliers**

   Extreme observations can affect the model.

3. **Multicollinearity**

   Highly correlated independent variables can make coefficient estimates unstable.

Disadvantages of Logistic Regression

4. **Feature Engineering May Be Required**

   For complex nonlinear relationships, transformations or additional features may be needed.

5. **Not Ideal for Very Complex Problems**

   Models such as decision trees, random forests, neural networks, etc. may perform better on complicated patterns.

6. **Requires Appropriate Data Preparation**

   Missing values, extreme outliers, and poorly scaled or encoded features can affect performance.

Types of Logistic Regression

A. Binary Logistic Regression

B. Multinomial Logistic Regression

C. Ordinal Logistic Regression

A. Binary Logistic Regression

Used when there are **two classes**.

Examples:

Pass / Fail

Yes / No

Spam / Not Spam

Types of Logistic Regression

B. Multinomial Logistic Regression

Used when there are **more than two classes that do not have a natural order**.

Example:

Predicting the type of transport:

Car

Bus

Train

Types of Logistic Regression

C. Ordinal Logistic Regression

Used when there are **more than two categories that have a natural order.**

Example:

Customer satisfaction:

Poor

Average

Good

Linear Regression vs Logistic Regression

Feature	Linear Regression	Logistic Regression
Main purpose	Predict numerical values	Predict classes
Output	Continuous value	Probability/class
Typical target	Salary, price, marks	Yes/No, Pass/Fail
Main function	Linear equation	Sigmoid/logistic function
Output range	Unrestricted	Probability: 0 to 1
Common loss	Squared error / least squares	Log loss
Example	Predict house price	Predict whether house sells
Common metrics	MAE, MSE, RMSE, R^2	Accuracy, Precision, Recall, F1

Classification Models

Classification is a **supervised machine learning technique** used to predict a **categorical (discrete) output** or **class label** based on **one or more input features**.

In simple word, Classification models **learn from labeled data** and **assign new data** to one or more **predefined categories** or **classes**.

For example:

- Email → **Spam / Not Spam**
- Transaction → **Fraudulent / Genuine**
- Patient → **Disease / No Disease**
- Student → **Pass / Fail**

Classification Models

suppose we want to predict whether a student will pass an examination.

Input features:

- ✓ Attendance Marks
- ✓ Assignment Marks
- ✓ Weekly Exam Marks
- ✓ Mid Sem Exam marks
- ✓ Theory Exam Marks

Output:

- ➢ Pass **OR** Fail

Classification Models

Major Classification Algorithms-

There are several commonly used classification algorithms.

1. Decision Tree
2. Naïve Bayes
3. SVM
4. Random Forest

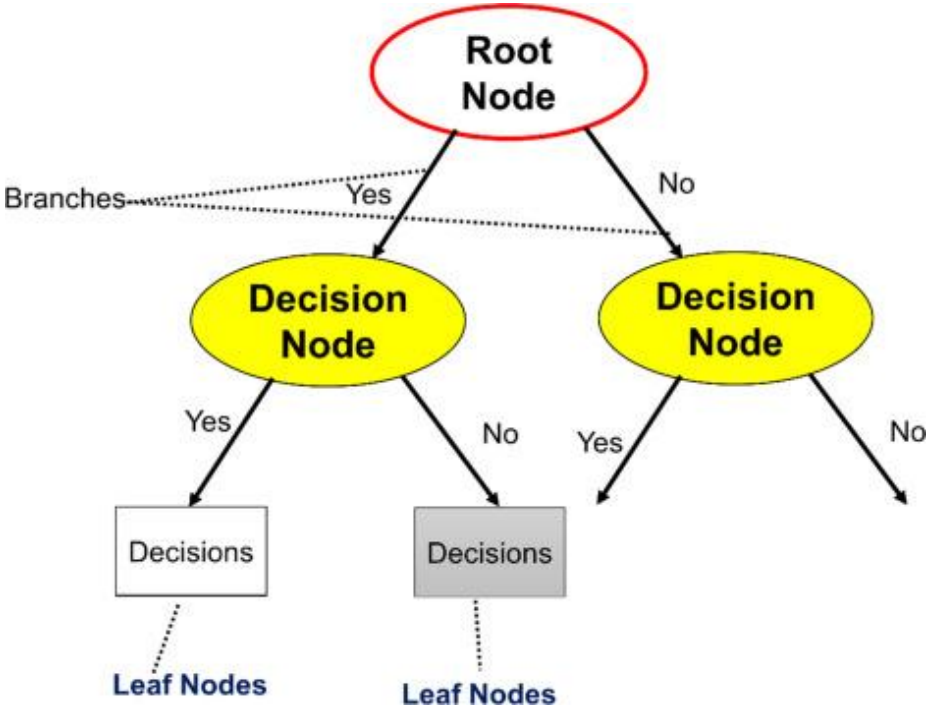
Decision Tree

A Decision Tree is a **supervised machine learning algorithm** used for both **classification** and **regression** problems. It makes predictions by repeatedly dividing the dataset based on **conditions** or **rules**.

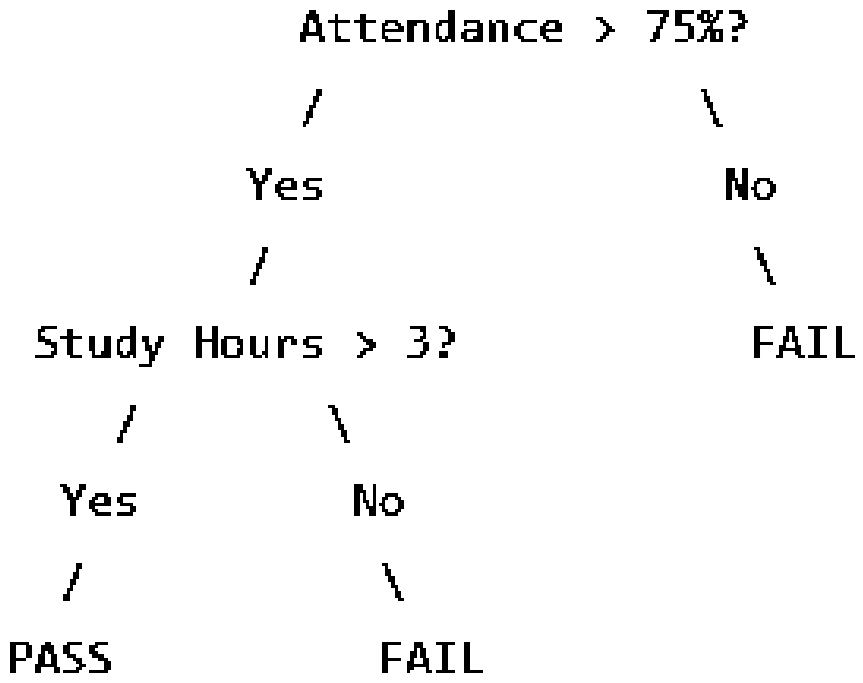
It looks like a tree structure, where **each internal node** represents a **decision**, **each branch** represents an **outcome**, and **each leaf node** represents the **final prediction**.

Decision Tree

Basic Structure



Example



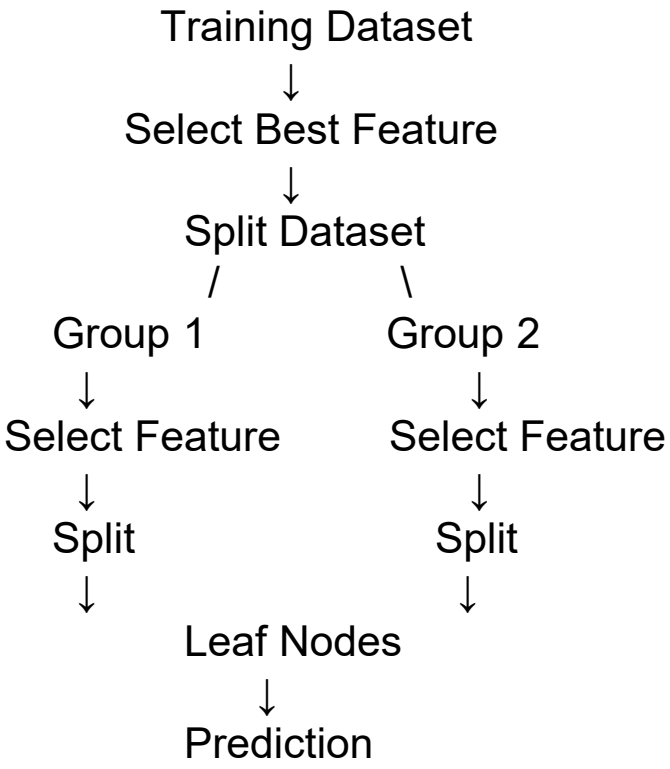
Decision Tree

Here:

- **Root Node:** Attendance > 75%?
- **Decision/Internal Nodes:** Study Hours > 3?
- **Branches:** Yes / No
- **Leaf Nodes:** PASS / FAIL

How Does a Decision Tree Work?

The Decision Tree starts with the complete dataset and repeatedly divides it into smaller groups.



The algorithm tries to select the **best feature** and **best split** at each stage.

Why Do We Use Decision Trees?

We use Decision Trees because they can make predictions using a series of simple **if-then decisions**.

For example - To predict whether a student will pass:

IF Attendance > 75%
↓
YES
↓
IF Study Hours > 3
↓
YES
↓
PASS

Instead of using complicated mathematical relationships, the Decision Tree represents the decision-making process in an easily understandable form.

Common application

Decision Trees are used for:

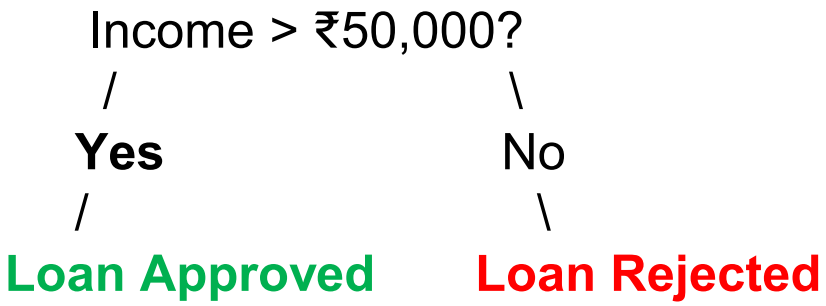
- ✓ **Student result prediction** → Pass/Fail
- ✓ **Loan approval** → Approved/Rejected
- ✓ **Medical diagnosis** → Disease/No Disease
- ✓ **Spam detection** → Spam/Not Spam
- ✓ **Customer churn** → Churn/Not Churn
- ✓ **Fraud detection** → Fraud/Genuine
- ✓ **Purchase prediction** → Buy/Don't Buy

Importance of Decision Trees

1. Easy to Understand

The biggest advantage is that the decision-making process can be represented as a tree.

For example:



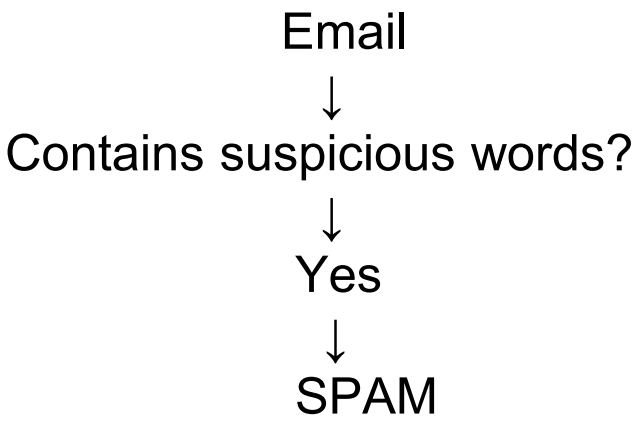
Even a non-technical person can understand this logic.

Importance of Decision Trees

2. Useful for Classification

Decision Trees are very effective for classification problems.

For example:



Importance of Decision Trees

3. Can Handle Nonlinear Relationships

A Decision Tree does not require the relationship between features and target to be linear.

For example - the decision may depend on multiple conditions:

Age > 30?
↓
Income > ₹60,000?
↓
Credit Score > 700?
↓
Loan Approved

Importance of Decision Trees

4. Requires Less Data Preprocessing

Decision Trees generally do not require feature scaling such as:

- Standardization
- Min-Max normalization

For example - features such as:

Age = 25

Income = 80,000

Credit Score = 750

can be used without putting them on the same numerical scale.

Importance of Decision Trees

5. Helps in Decision-Making

The tree provides a clear sequence of decisions, making it useful when understanding **why** a particular prediction was made.

Advantages of Decision Trees

1. **Easy to Understand**

   The tree structure is intuitive and can be explained using simple rules.

2. **Easy to Visualize**

   Decision Trees can be displayed graphically.

3. **Handles Classification and Regression**

   The same basic approach can be used for both types of problems.

4. **Little Data Preprocessing**

   Feature scaling is generally not required.

Advantages of Decision Trees

5. Handles Numerical and Categorical Features

Decision Trees can work with different types of input features, although implementation-specific preprocessing may be needed for categorical variables.

6. Captures Nonlinear Relationships

They can model complex decision boundaries without requiring a linear relationship.

7. Feature Selection

The tree naturally selects useful features while creating splits.

8. Provides Interpretable Rules

Disadvantages of Decision Trees

1. Overfitting

A Decision Tree can become too deep and memorize the training data.

For example:

Training Accuracy = 100%

Testing Accuracy = 70%

This indicates possible overfitting.

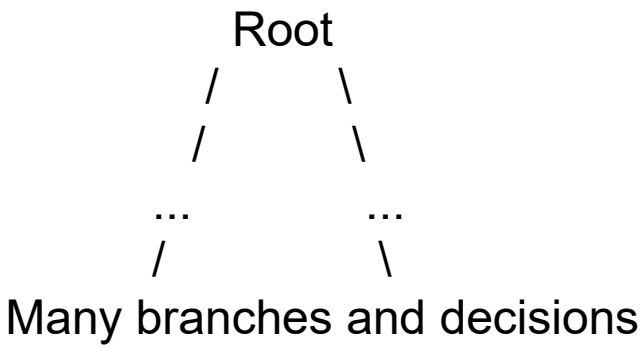
Disadvantages of Decision Trees

2. Unstable

Small changes in the training data can sometimes result in a significantly different tree.

3. Deep Trees Become Complex

A very large tree can be difficult to understand.



Disadvantages of Decision Trees

4. Can Be Biased Toward Certain Splits

Depending on the splitting criterion and data, trees can favor features with many possible split points or categories.

5. Greedy Learning

The tree usually chooses the best split at the current node rather than searching all possible future trees for a globally optimal structure.

6. Poor Generalization Without Control

Techniques such as maximum depth, minimum samples, pruning per leaf can help control tree complexity.

Random Forest

Random Forest is a **supervised machine learning algorithm** that **combines multiple Decision Trees** to make a more accurate and reliable prediction.

It creates a collection (**forest**) of many decision trees and each tree is built using random samples of the data and typically a random subset of features.

Random Forest can be used for both:

Classification → predicting categories / classes

Regression → predicting numerical values

Random Forest

For example:

- ✓ Spam / Not Spam
- ✓ Pass / Fail
- ✓ Fraud / Genuine
- ✓ Disease / No Disease
- ✓ House Price prediction

Why Do We Use Random Forest?

The main reason for using Random Forest is to overcome some of the limitations of a **single Decision Tree**.

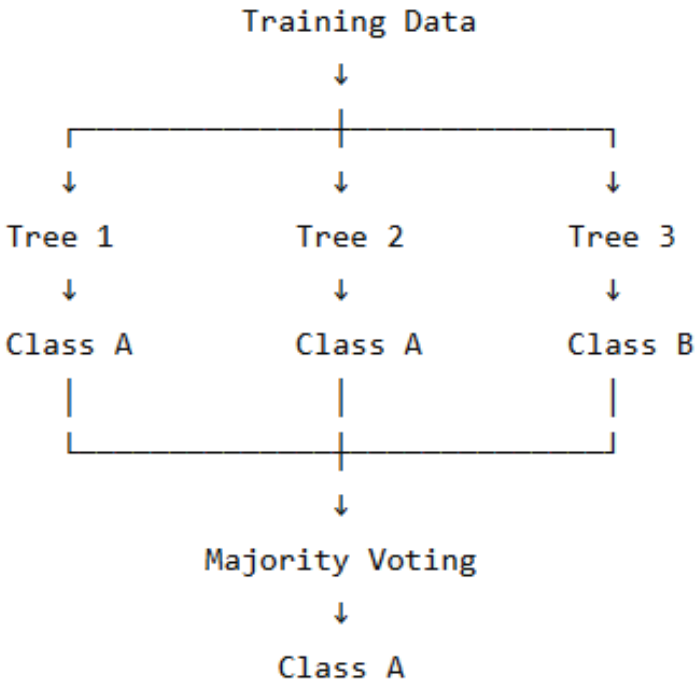
A single Decision Tree can easily **overfit** the training data.

For example -

Decision Tree
↓
Training Data
↓
Very complex tree
↓
Overfitting

Random Forest creates many different trees:

Why Do We Use Random Forest?



Because many trees participate in the final decision, Random Forest generally produces a **more stable and robust prediction** than a single decision tree.

How Does Random Forest Work?

Random Forest mainly follows these steps:

Step 1: Select Random Samples

The algorithm takes different random samples from the training dataset. This process is called **Bootstrap Sampling** or **Bagging**.

How Does Random Forest Work?

Random Forest mainly follows these steps:

Step 2: Build Multiple Decision Trees

A separate Decision Tree is constructed using each bootstrap sample.

Dataset

↓

Random Sample 1 → Tree 1

Random Sample 2 → Tree 2

Random Sample 3 → Tree 3

Random Sample 4 → Tree 4

Random Sample 5 → Tree 5

How Does Random Forest Work?

Random Forest mainly follows these steps:

Step 3: Random Feature Selection

At each split, the tree considers only a random subset of available features. This makes the trees different from each other and helps reduce correlation among them.

How Does Random Forest Work?

Random Forest mainly follows these steps:

Step 4: Each Tree Makes a Prediction

Suppose we have 5 trees –

Tree 1 → Pass

Tree 2 → Pass

Tree 3 → Fail

Tree 4 → Pass

Tree 5 → Pass

How Does Random Forest Work?

Random Forest mainly follows these steps:

Step 5: Final Prediction

For classification, the Random Forest generally uses majority voting.

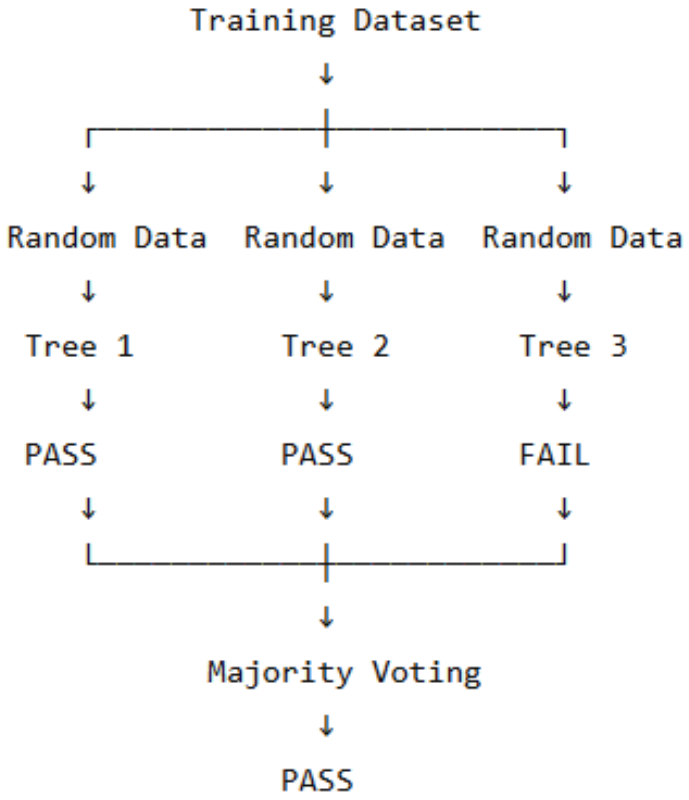
Pass = 4 votes

Fail = 1 vote

Final Prediction = PASS

For regression, the predictions of the trees are generally **averaged**.

Simple Random Forest Structure



A real Random Forest can contain dozens or hundreds of trees.

Importance of Random Forest

Random Forest is important because it provides a good balance between **prediction accuracy**, **robustness** and **flexibility**.

1. **Reduces Overfitting**

   A single Decision Tree may learn the training data too closely.

   Random Forest combines many trees, which generally reduces the effect of an individual tree's errors.

2. **Improves Prediction Accuracy**

   Multiple trees contribute to the final decision.

   Therefore, the final prediction can be more reliable than the prediction from one tree.

Importance of Random Forest

3. **Handles Complex Relationships**

   Random Forest can capture nonlinear relationships between input features and the target.

4. **Useful for Large and Complex Datasets**

   Random Forest can work well when a dataset contains many observations and features.

5. **Provides Feature Importance**

   Random Forest can estimate which features are useful for prediction.

Advantages of Random Forest

1. High Prediction Accuracy

Random Forest often provides strong performance on structured/tabular data.

2. Less Overfitting Than a Single Decision Tree

Because many trees are combined, the model is generally more robust than one unrestricted tree.

3. Handles Nonlinear Data

It does not require a linear relationship between features and target.

Advantages of Random Forest

4. No Feature Scaling Required

Unlike algorithms such as KNN or SVM, Random Forest generally does not require stand.

5. Handles Many Features

Random Forest can work with a large number of input variables.

6. Provides Feature Importance

It can help identify important features.

7. Robust to Noise

Because predictions are aggregated across many trees, the model can be relatively robust to noisy observations.

Disadvantages of Random Forest

1. Computationally More Expensive

Since many Decision Trees must be created, Random Forest generally requires more computation than a single Decision Tree.

2. Requires More Memory

Storing many trees requires more memory.

3. Less Interpretable

A single Decision Tree can easily be visualized.

Disadvantages of Random Forest

4. Prediction Can Be Slower

When there are many trees, each tree must make a prediction before the final result is calculated.

5. Can Still Overfit

Although Random Forest generally reduces overfitting, it is not completely immune to it, especially with unsuitable model settings or noisy/high-dimensional data.

Disadvantages of Random Forest

5. Hyperparameter Selection

Performance can depend on parameters such as:

- Number of trees
- Maximum depth

Naïve Bayes

Naïve Bayes is **supervised machine learning algorithms** used mainly for **classification**. It works in very different way.

Naïve Bayes is a **probabilistic classification algorithm** based on **Bayes' Theorem**.

Naïve Bayes

For example, for an email:

Email
↓
Contains "Offer", "Free", "Prize"
↓
Naïve Bayes
↓
P(Spam) = 0.92
P(Not Spam) = 0.08
↓
Prediction = SPAM

Why Do We Use Naïve Bayes?

Naïve Bayes is particularly useful when:

1. The dataset contains a **large number of features**.
2. We need **fast training** and **prediction**.
3. The data is **text-based**.
4. We need a simple probabilistic model.
5. We want a good baseline classification model.

Common applications

- ✓ Spam email detection
- ✓ Sentiment analysis
- ✓ News classification
- ✓ Document classification
- ✓ Text categorization
- ✓ Language identification
- ✓ Medical diagnosis
- ✓ Customer classification

How Does Naïve Bayes Work?

Suppose we want to classify an email as: **Spam** or **Not Spam**

The email contains: **"Free Prize Offer“**

Naïve Bayes calculates the probability of both classes based on the observed words.

Types of Naïve Bayes

There are several variants.

1. **Gaussian Naïve Bayes**

   Used when features are continuous and are modeled using a Gaussian distribution.

   Example: Age, Height, Weight, Blood Pressure

2. **Multinomial Naïve Bayes**

   Commonly used for **text classification**, especially when features represent counts or frequencies.

   Example: Word counts in an email

Types of Naïve Bayes

3. Bernoulli Naïve Bayes

Used when features are binary.

Example: Contains "free" → 1

Contains "offer" → 1

Contains "loan" → 0

Importance of Naïve Bayes

1. **Very Fast**

   It is computationally efficient and can train quickly even on large datasets.

2. **Excellent for Text Classification**

   It is widely used for - Email → Spam/Not Spam,

   Review → Positive/Negative, News → Sports/Politics/Technology

3. **Works with High-Dimensional Data**

   Text datasets may contain thousands of words/features. Naïve Bayes can handle such datasets effectively.

Importance of Naïve Bayes

4. Provides Probabilistic Predictions

It can provide probabilities associated with different classes, which can be useful for decision-making.

Advantages of Naïve Bayes

1. Simple to implement
2. Fast training
3. Fast prediction
4. Works well with high-dimensional data
5. Particularly effective for text classification
6. Can work with relatively small training datasets
7. Supports probabilistic predictions
8. Requires comparatively low computational resources

Disadvantages of Naïve Bayes

- **Independence Assumption** – Assumes features are independent, which may not be true.
- **Zero-Frequency Problem** – A missing feature can make the prediction zero.
- **Correlated Features** – Performance may decrease when features are closely related.
- **Limited Relationships** – Cannot easily handle complex relationships between features.
- **Probability Issues** – Predicted probabilities may not always be accurate.

SVM — Support Vector Machine

Support Vector Machine (SVM) is a **supervised machine learning algorithm** primarily used for classification.

The main idea is to find a **decision boundary (hyperplane)** that separates different classes while maximizing the **margin** between them.

Why Do We Use SVM?

SVM is useful when:

- ✓ The classes can be separated clearly.
- ✓ The dataset has many features.
- ✓ We need a strong classification boundary.
- ✓ The dataset is relatively small or medium-sized.
- ✓ The relationship between features and classes may be nonlinear.

Why Do We Use SVM?

SVM is commonly used in:

- Image classification
- Text classification
- Face recognition
- Handwriting recognition
- Bioinformatics
- Medical classification
- Pattern recognition

Importance of SVM

1. **Effective in High-Dimensional Spaces**

   SVM can work well when there are many features.

2. **Maximum-Margin Classification**

   SVM explicitly focuses on finding a boundary with a large margin.

3. **Handles Nonlinear Classification**

   Kernel functions allow SVM to model nonlinear decision boundaries.

4. **Strong Generalization**

   SVM can generalize well to unseen data.

Importance of SVM

5. Useful for Small and Medium Datasets

SVM can be particularly effective when the dataset is not extremely large but contains meaningful features.

Advantages of SVM

- ✓ Effective in high-dimensional datasets.
- ✓ Can work well with relatively small and medium-sized datasets.
- ✓ Provides a clear mathematical decision boundary
- ✓ Kernel functions can handle nonlinear relationships.
- ✓ Uses support vectors, making the final boundary dependent on a subset of training observations.
- ✓ Often provides good generalization when properly tuned.

Disadvantages of SVM

1. **Computationally Expensive for Large Datasets**

   Training can become slow when the number of observations is very large.

2. **Parameter Selection is Important**

   Poor parameter selection can significantly affect performance.

3. **Feature Scaling is Usually Important**

   SVM is sensitive to feature scales.

4. **Difficult to Interpret**

   Compared with a Decision Tree, an SVM model is generally harder to explain in terms of simple human-readable rules.

Disadvantages of SVM

5. Kernel Selection Can Be Difficult

Choosing between: Linear, Polynomial, RBF, Sigmoid depends on the dataset and problem.

Clustering Models

Clustering is an **unsupervised machine learning technique** used to group similar data points into **clusters** without having predefined class labels.

In classification, we already know the classes:

Student → Pass / Fail

But in clustering, the classes are **not given**. The algorithm discovers groups based on similarity.

Clustering Models

Simple Example

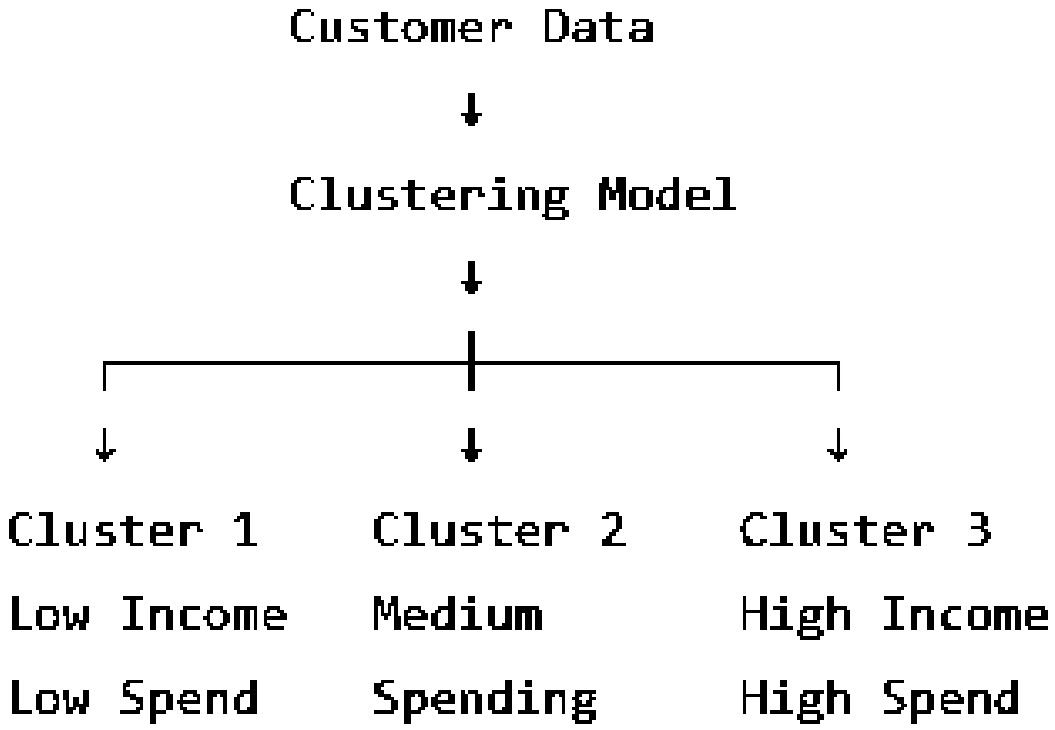
Suppose we have customers with information about:

- ✓ Age
- ✓ Income
- ✓ Spending Score

We may not know the customer categories beforehand.

A clustering algorithm can discover groups such as -

Clustering Models



These clusters can then be interpreted as customer segments.

Why Do We Use Clustering?

Clustering is used when we have **unlabeled data** and want to discover **hidden patterns** or **natural groups**.

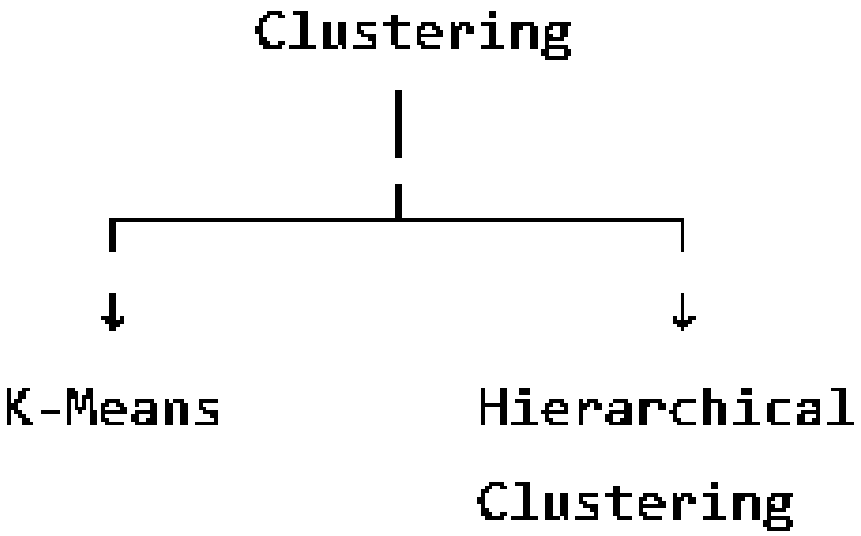
Common applications

1. Customer segmentation
2. Image segmentation
3. Document grouping
4. Social network analysis
5. Recommendation systems
6. Gene/data analysis
7. Geographic analysis

Why Do We Use Clustering?

Main Clustering Models

Two important clustering techniques are:



K-Means

K-Means is an **unsupervised machine learning algorithm** that divides data into **K clusters** based on similarity.

Here, **K** represents the number of clusters we want to create.

For example:

K = 3

means we want the algorithm to create **3 clusters**.

Why Do We Use K-Means?

K-Means is useful when we want to:

- Divide customers into groups.
- Identify similar products.
- Group documents.
- Segment images.
- Discover patterns in unlabeled data.
- Simplify a large dataset into meaningful groups.

Importance of K-Means

1. Customer Segmentation

Companies can divide customers into groups based on behavior.

2. Data Exploration

It helps discover hidden patterns in unlabeled data.

3. Image Segmentation

Pixels with similar characteristics can be grouped together.

4. Document Clustering

Similar documents can be grouped automatically.

Importance of K-Means

5. Data Compression

K-Means can represent many observations using a smaller number of cluster centroids.

6. Pattern Recognition

It can identify natural groups within datasets.

Advantages of K-Means

1. **Simple and Easy to Understand**

   The algorithm is relatively straightforward.

2. **Easy to Implement**

   It is available in many machine learning libraries.

3. **Computationally Efficient**

   K-Means can be relatively fast, particularly for datasets where the number of observations and dimensions is manageable.

4. **Works Well for Compact Clusters**

   It works particularly well when clusters are reasonably compact and separated.

Advantages of K-Means

5. Scalable

K-Means can be applied to relatively large datasets.

6. Easy to Interpret

The resulting clusters and their centroids can often be analyzed easily.

Disadvantages of K-Means

1. **Need to Choose K**

   The number of clusters generally needs to be specified beforehand.

2. **Sensitive to Initial Centroids**

   Different starting centroids can sometimes produce different results.

   Methods such as **k-means++** are commonly used to obtain better initial centroids.

3. **Sensitive to Outliers**

   Extreme observations can significantly affect cluster centroids.

Disadvantages of K-Means

4. Works Best with Certain Cluster Shapes

Standard K-Means generally works best when clusters are relatively compact and roughly spherical.

It may struggle with irregularly shaped clusters.

5. Feature Scaling Can Be Important

If features have very different scales, large-scale features can dominate distance calculations.

Hierarchical Clustering

Hierarchical Clustering is an **unsupervised learning technique** that creates a **hierarchy of clusters**.

Instead of directly specifying only one final set of clusters, it builds a **tree-like structure** showing how observations or groups are progressively merged or divided.

The most common approach is **Agglomerative Hierarchical Clustering**.

Example of Hierarchical Clustering

Suppose we have customers:

Customer A

Customer B

Customer C

Customer D

Customer E

Based on purchasing behavior:

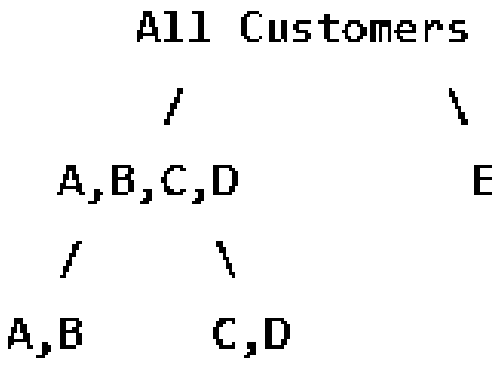
A and B → Very Similar

C and D → Very Similar

E → Different

Example of Hierarchical Clustering

Hierarchical clustering may produce:



So we can select:

Cluster 1 → A, B

Cluster 2 → C, D

Cluster 3 → E

Importance of Hierarchical Clustering

1. Shows Relationships

It provides information about how observations are related at different levels of similarity.

2. Does Not Require K at the Start

Unlike standard K-Means, hierarchical clustering can construct the hierarchy without first specifying a single final number of clusters.

3. Dendrogram Visualization

The dendrogram makes the hierarchical relationships easier to visualize.

Importance of Hierarchical Clustering

4. Useful for Small and Medium Datasets

It can be particularly useful when understanding the structure of a dataset is more important than scalability.

5. Useful in Biological Analysis

6. Useful for Customer and Document Grouping

Similar customers or documents can be organized into hierarchical groups.

Advantages of Hierarchical Clustering

1. No Need to Specify K Initially

The algorithm builds a hierarchy, and a desired number of groups can be selected afterward.

2. Dendrogram is Easy to Visualize

The dendrogram provides a graphical representation of cluster relationships.

3. Shows Multiple Levels of Clustering

You can examine broad groups as well as smaller subgroups.

4. Flexible Linkage Choices

Different linkage methods can be selected according to the characteristics of the data.

Advantages of Hierarchical Clustering

5. Useful for Exploratory Analysis

It can reveal relationships that may not be obvious from the raw dataset.

Disadvantages of Hierarchical Clustering

1. Computationally Expensive

It can become expensive for very large datasets.

2. Sensitive to Distance and Linkage

The results can change depending on:

- Distance metric
- Linkage method

3. Sensitive to Outliers

Outliers can affect the structure of the hierarchy.

Disadvantages of Hierarchical Clustering

4. Merges Are Generally Irreversible

In standard agglomerative clustering, once two clusters are merged, that merge is not normally undone.

5. Difficult to Scale to Very Large Datasets

For very large datasets, K-Means or other scalable clustering approaches may be more practical.

Reference Books

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